Course Name: Engineering Chemistry
Course Code: CY-101
Course Type: Core

Contact Hours/Week: 3L **Course Credits:** 03

Course Objectives

- To familiarize the students about the chemistry of engineering materials and applications.
- To enable the students to understand about the fundamentals of characterization techniques for different materials.
- To introduce equilibrium states of various systems with help of phases, components etc.
- To develop an understanding of electrochemistry and its applications in fuel cell/batteries and controlling corrosion.
- To familiarize the students about green chemistry and applications.

Unit Number	Course Content	Contact Hours
UNIT-01	**Engineering Materials:** Introduction to Engineering materials-Semiconductors- conducting polymers, Cement, Liquid crystals, Composite materials, Nanomaterials	05L
UNIT-02	**Characterization Techniques-I:** Introduction to spectroscopy, UV-Visible spectroscopy-Absorption laws, Instrumentation, formation of absorption bands, Chromophore and auxochrome concept, application of UV-Visible spectroscopy; IR spectroscopy - Principle, selection rules, spectral features of some classes of compounds, important features of IR spectroscopy NMR spectroscopy-Principle, relaxation processes, instrumentation, shielding-deshielding effects, spin-spin coupling, coupling constant, applications of NMR	08L
UNIT-03	**Characterization Techniques-II:** Introduction to Thermal methods, instrumentation and applications (TGA, DTA, DSC). Introduction to microscopy techniques- instrumentation and applications (AFM and SEM).	04L
UNIT-04	**Phase rule:** Introduction to phase rule, one component system, two component systems, advantages and limitations.	05L
UNIT-05	**Electrochemistry:** General concepts of electrochemistry, concentrations cell, galvanic cell, Fuel cells-hydrogen-oxygen fuel cell, solid oxide fuel cells, Batteries-dry cell, lead storage, Nickel-cadmium, Li-ion batteries, functioning and their applications. Applications of electrochemistry in corrosion and its control: Introduction, Types of corrosion-chemical and electrochemical, Mechanisms of corrosion, factors affecting corrosion and different protection methods for corrosion control.	10L
UNIT-06	**Green Chemistry:**Introduction to green chemistry, Twelve Principles of green chemistry, Synthesis of biofuel, Alternative sources of energy: Photo-catalysis, Solar cells.	04L

Course Outcomes

Upon successful completion of the course, the students will be able to:

CO1: Understand basic concepts of engineering materials and applications in various fields
CO2: Identify instrumental techniques for analysis of different materials
CO3: Understand the behavior of systems with changes in variables
CO4: Understand the electrochemistry, batteries, fuel cells, engineering problems related to corrosion and to achieve a practical solution
CO5: Understand basic concepts of green chemistry their applications in various fields

Text books

1. Engineering Chemistry by Shashi Chawla, Dhanpat Rai & Co.
2. Engineering Chemistry by S. Vairam and S. Ramesh, Wiley.
3. Engineering Chemistry by Jain & Jain, Dhanpat Rai Publishing Co.

Reference books

1. Materials Science and Engineering: An Introduction Hardcover by William D. Callister, 6th Edition, John Wiley & Sons.
2. Physical Chemistry, by Peter Atkins, Julio de Paula, James Keeler, International Edition, 2018, Oxford University Press.